Fahim Ahamed

Authorized to work in the U.S. and do not require sponsorship - U.S. Permanent Resident
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Summary

Data Scientist with a background in **AI research** and hands-on experience across healthcare, NLP, and geoscience. Builds **end-to-end ML pipelines**, conducts statistical and spatial analysis, and trains models for real-world research applications. Develops **agentic AI pipelines** using RAG, LLMs, and MCP. Published **peer-reviewed research** across IEEE, SAGE, and Springer Nature. Comfortable working from raw data to results and collaborating directly with domain experts. Proficient in Python, SQL, R, PyTorch, and TensorFlow.

Education

The City College of New York — New York, NY
M.S. in Data Science and Engineering | GPA: 3.83/4.0 — *Dec 2027*

East West University — Dhaka, Bangladesh
B.S. in Computer Science and Engineering | GPA: 3.74/4.0 — *Dec 2023*

Technical Skills

Languages: Python, SQL, R
Data Analysis: NumPy, Pandas, SciPy, Excel
Data Visualization: Matplotlib, Seaborn, Plotly, ggplot2, Tableau
AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, OpenCV, NLTK, Word2Vec
LLMs & Agents: LLMs, RAG, MCP, Hugging Face, ChromaDB
Database & Big Data: MySQL, MongoDB, Snowflake
MLOps & Cloud: AWS, MLflow, Git/GitHub, Docker, Vercel

Work Experience

Data Core Intern (Healthcare Analytics) — Oct 2026 - May 2027
NYC Health + Hospitals — *New York, NY*

- Engineer and validate LLM prompts to evaluate **commercial LLMs' capability to automate population health analytics** tasks like **health disparity detection** and report generation.
- Run **blinded, human-in-the-loop evaluations** of LLM outputs, measuring accuracy, bias, and **hallucination risk** for responsible use in a safety net health system.
- Translate evaluation results into clear findings for the research team, supporting **manuscript development** and **grant dissemination** on responsible LLM use in healthcare.

Data Science Intern — Oct 2026 - Dec 2026
Tactorum — *New York, NY*

- Develop **deep-learning pose-estimation models** (PyTorch) that track body and facial keypoints in high-speed video, automating **pain behavior measurement** in preclinical research.
- Own the data side of model training, curating and validating **labeled datasets** with rigorous **quality review** to ensure model generalization across subjects and labs.
- Run and benchmark **training experiments** on **cloud platforms**, defining evaluation metrics and diagnosing failure modes to improve reliability.

Research Assistant — Dec 2025 - Present
CUNY Research Foundation — *New York, NY*

- Develop an **agentic AI pipeline** using RAG, LLMs, and MCP-connected tools for natural language querying over complex geoscience databases, as part of an **NSF-funded** geoinformatics project.
- Bridge **geoscience domain expertise** and **AI engineering**, translating research requirements into technical specs and iterating on pipeline outputs for interpretability and scientific relevance.
- Design the **end-to-end technical stack**: **vector database architecture**, domain-specific embedding models, LLM optimization, and cloud deployment for a production-ready geoscience research tool.

Data Analyst Oct 2025 - Feb 2026, Jul 2026 - Present

The City College of New York *New York, NY*

- Analyze **institutional data** using Excel, Tableau, and Python, surfacing patterns that shape **advising and student engagement programs**.
- Conduct **data-driven surveys** and **retention studies**, uncovering behavioral patterns that inform targeted outreach.
- Build **analytical reports** and **dashboards** that surface trends and anomalies across departments, replacing gut-feel decisions with data-backed ones.

Data Evangelist Intern Feb 2026 - May 2026

DataCamp *New York, NY*

- Designed and maintained **scalable data pipelines** to collect, integrate, and validate **multi-source data** through automated scripts, APIs, and web scraping.
- Built **data applications** and analytical workflows powered by **LLMs**, integrating OpenAI and Hugging Face models into user-facing tools for exploring key metrics.
- Analyzed data to surface trends and key drivers, packaging findings into **visualizations** and structured reports.

Projects

NYC HIV/AIDS Trend Analysis & Predictive Modeling | *R, Tidyverse, ggplot2, sf, rpart, RandomForest, caret*

- Cleaned and engineered multi-year NYC Open Data on HIV/AIDS diagnoses, resolving **suppressed values** withheld for privacy, into analysis-ready datasets.
- Performed **spatiotemporal analysis** revealing persistent geographic and demographic disparities, and applied **Random Forest models ($\mathbf{R^2 = 0.65}$)** to identify high-impact predictors for public health targeting.

NYC 911 Mental Health Dispatch Analysis | *Python, Pandas, GeoPandas, Plotly, HTML/CSS, JavaScript*

- Built a **reproducible ETL pipeline** over **2.59M NYC 911 dispatch records**, deduplicating to a clean crisis-incident dataset and joining census blocks to precincts.
- Applied **per-capita normalization** and temporal decomposition across **78 precincts** to separate a real decline from a shift in dispatch practice, published as an interactive Plotly visualization.

Cursive Handwritten Prescription Recognition | *Python, TensorFlow, Keras, OpenCV, NumPy, Pandas, Matplotlib*

- Designed a **custom CNN architecture** and **modular preprocessing pipeline** (noise reduction, segmentation, character recognition) to digitize mixed Bangla-English prescriptions across varied cursive handwriting styles.
- Addressed variability in handwriting style and image quality through architecture refinement and data augmentation, then tuned hyperparameters to reach **91% median OCR accuracy**.

Research Experience

Explainable AI-Driven Hybrid Transfer Learning Framework for Accurate Skin Cancer Diagnosis – *Digital Health (SAGE)*

- Built a hybrid **EfficientNetB0 + Random Forest** model with advanced preprocessing for multi-class skin cancer classification, achieving **F1 98.64%** validated across datasets.

Robust Dual-Site Cancer Screening via Multi-Scale Vision Transformer and Rapid Recognition Pipeline – *IEEE Xplore*

- Developed a multi-scale Vision Transformer framework (Multi-ViT, MA-Transformer V2, MobileUNETR, TinyViT, Xception) for dual-site cancer screening, reaching **99.12% accuracy** on PAD-UFES-20.

Texture Feature-Based Colonic Polyp Detection Using Deep Learning Techniques – *Springer Nature*

- Segmented colonic polyps using texture features and **DeepLabv3+**, outperforming VGG16 and prior studies with **97.33% IoU** and setting a new benchmark in colorectal cancer detection.

Optimizing Energy Usage: Genetic Algorithms Leading the Charge Toward Sustainability – *Springer Nature*

- Applied **Genetic Algorithms** to household energy optimization, cutting annual electricity usage by **34.54%** with strong potential for nationwide scalability.

Accepted Papers (In Press)

A Lightweight Transformer Ensemble for Explainable Depression Emotion and Severity Detection – *IEEE (2025)*

- Designed a lightweight transformer ensemble with **LIME-based explanations** and a real-time interface for depression detection, achieving an **F1 of 81.63%**.